1.03 Complex Numbers: Argand Diagrams

Worked Solutions

1)

$A : 2 + 2i$

$B : 6 + 5i$

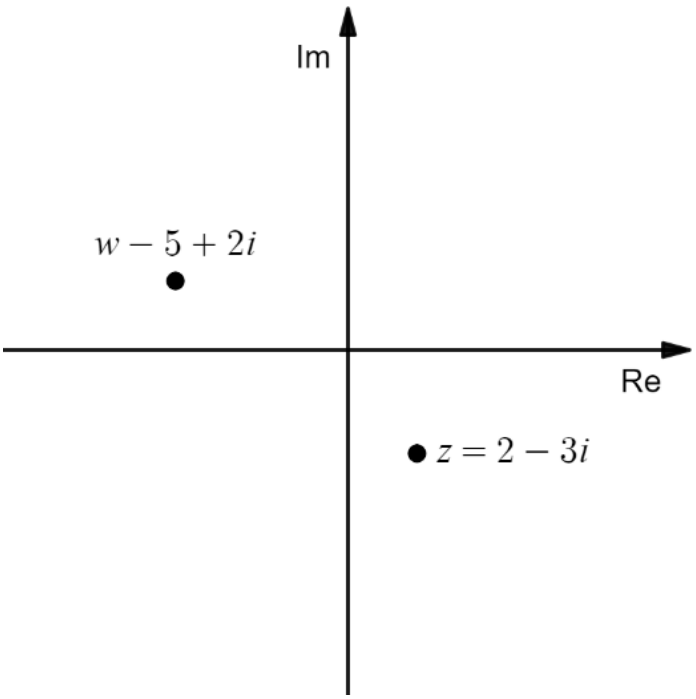
$C : -4 + 2i$

$D : -4 - 2i$

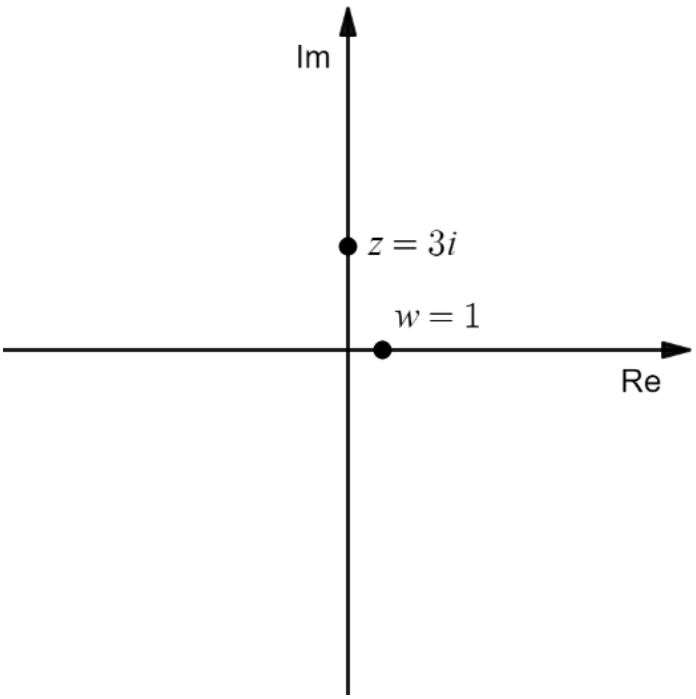
$E : -9 + 6i$

$F : 6 - 4i$

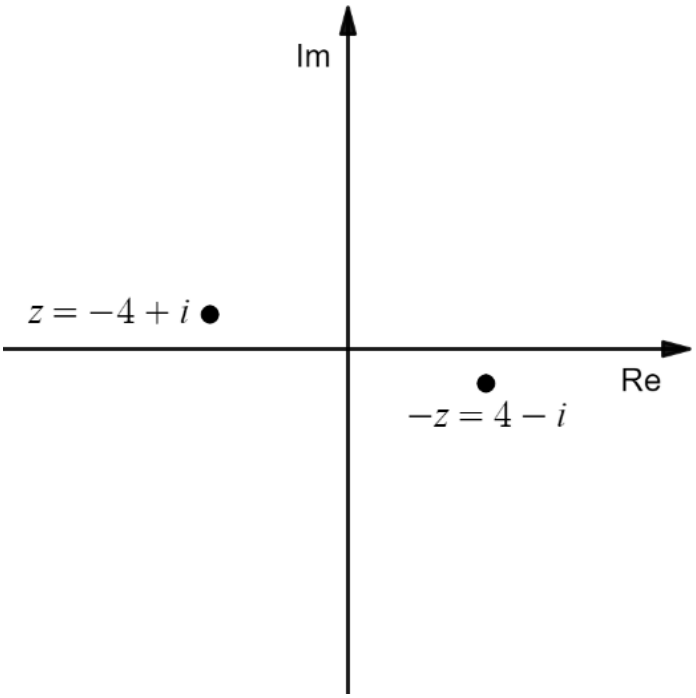
2 a)



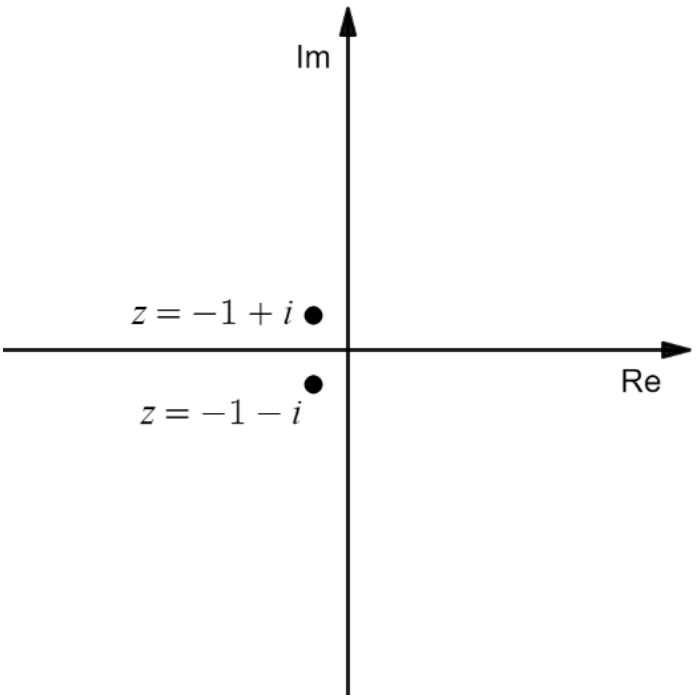
2 b)



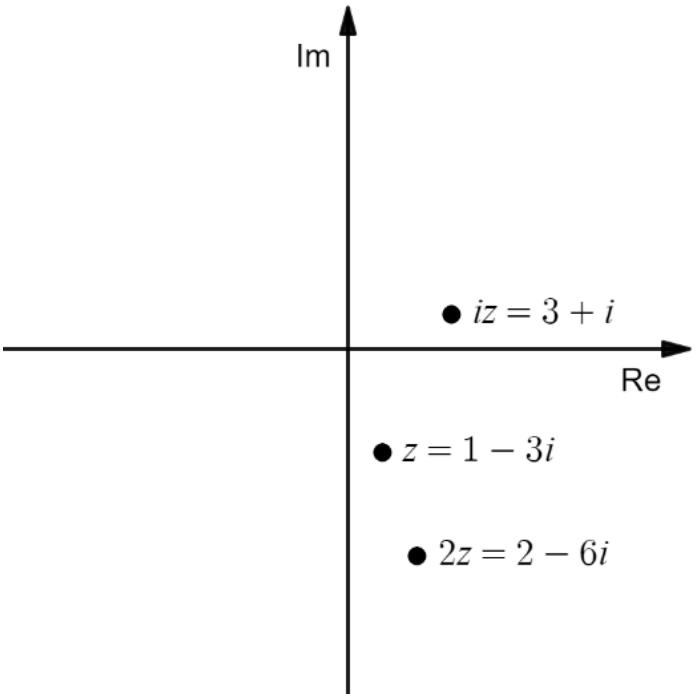
2 c)



2 d)

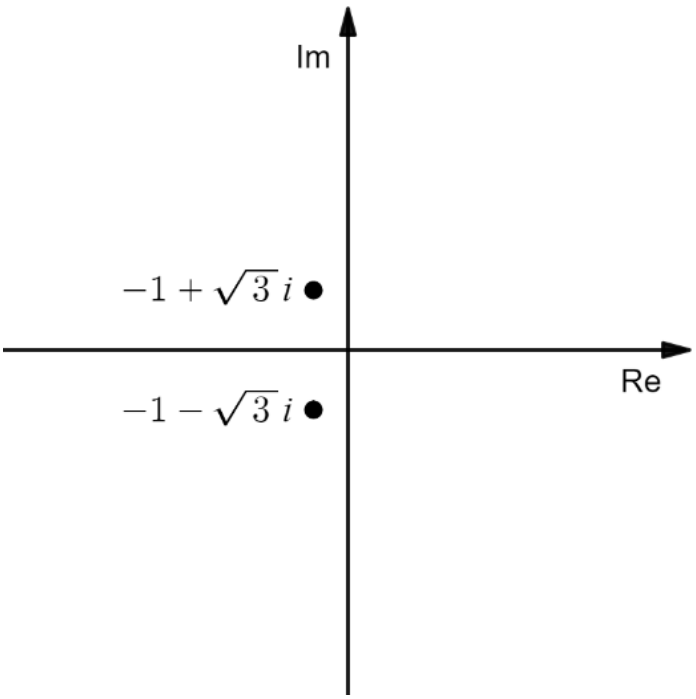


2 e)



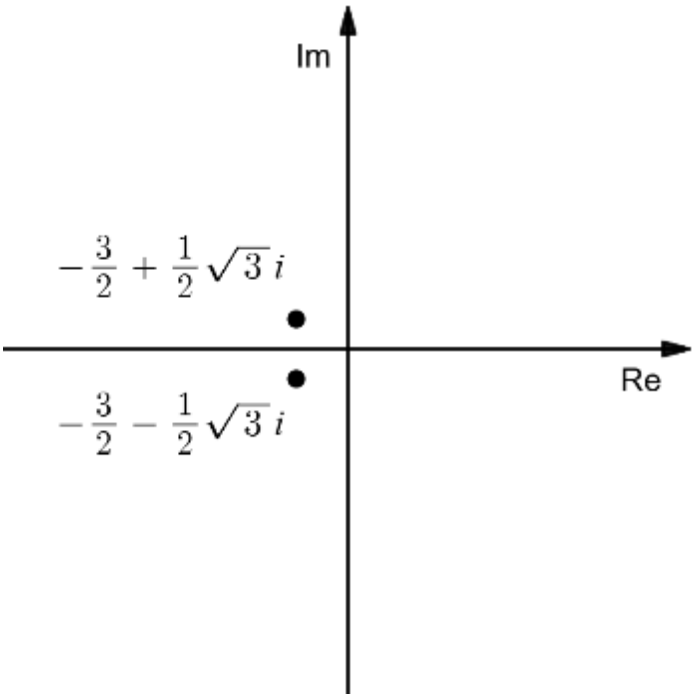
3 a)

$z^2 + 2z + 4 = 0$ has solutions $z = -1 \pm \sqrt{3}i$



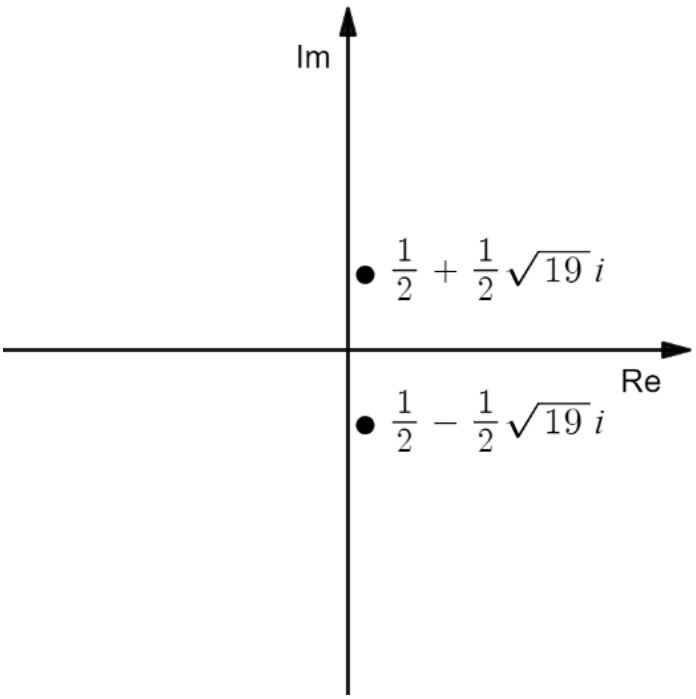
3 b)

$z^2 - 3z + 3 = 0$ has solutions $z = \frac{3}{2} \pm \frac{1}{2}\sqrt{3}i$



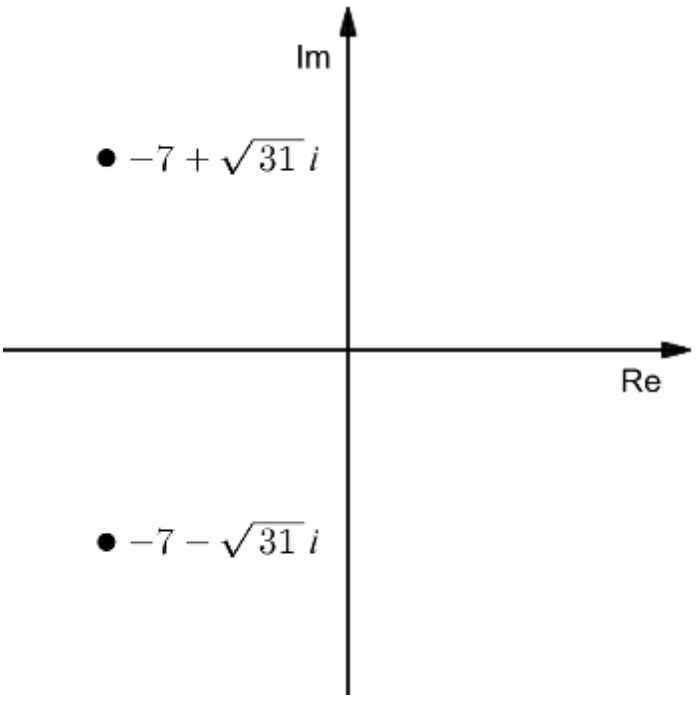
3 c)

$z^2 - z + 5 = 0$ has solutions $z = \frac{1}{2} \pm \frac{1}{2}\sqrt{19}i$



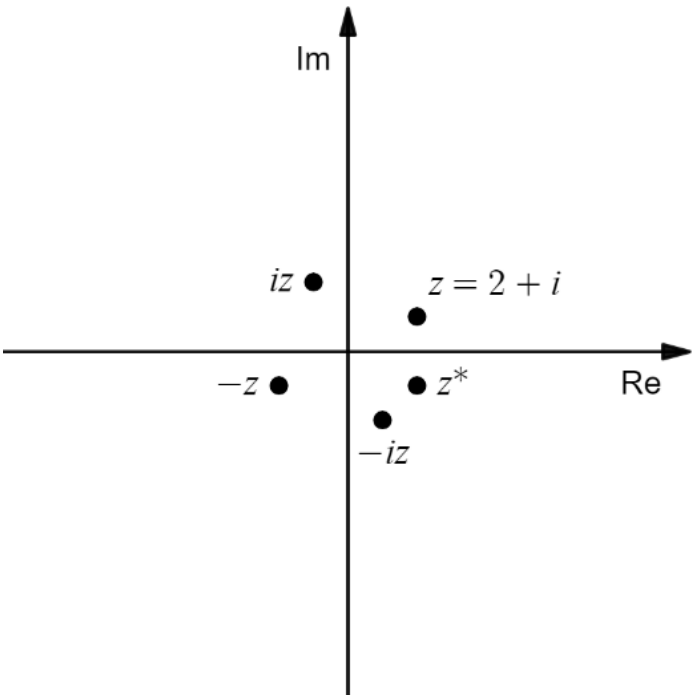
3 d)

$z^2 + 14z + 80 = 0$ has solutions $z = -7 \pm \sqrt{31}i$



4)

$z = 2 + i$, so that $z^* = 2 - i$, $-z = -2 - i$, $iz = -1 + 2i$, $-iz = 1 - 2i$



4 a)

$$z^* = 2 - i$$

Reflection across the real axis

4 b)

$$-z = -2 - i$$

Rotation by 180° about the origin

4 c)

$$iz = -1 + 2i$$

Rotation by 90° anticlockwise about the origin (or 270° clockwise)

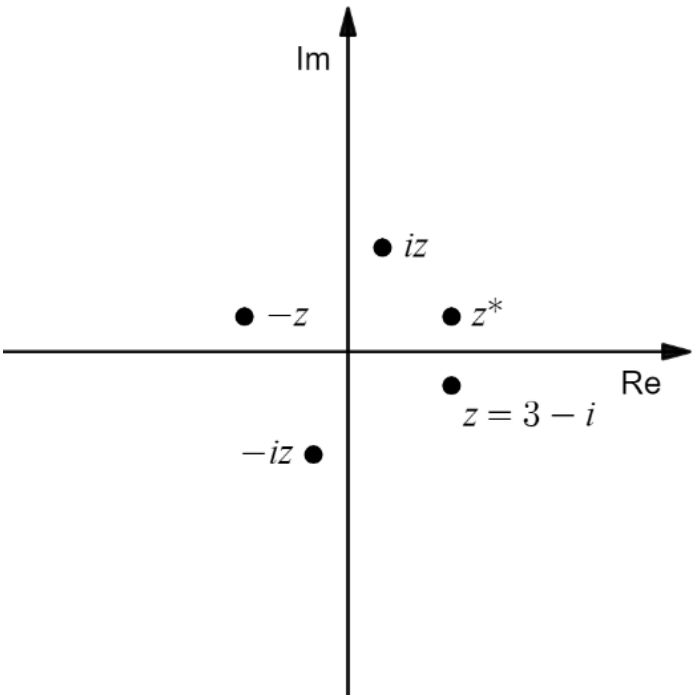
4 d)

$$-iz = 1 - 2i$$

Rotation by 270° anticlockwise about the origin (or 90° clockwise)

5)

$z = 3 - i$, so that $z^* = 3 + i$, $-z = -3 + i$, $iz = 1 + 3i$, $-iz = -1 - 3i$



5 a)

$$z^* = 3 + i$$

Reflection across the real axis

5 b)

$$-z = -3 + i$$

Rotation by 180° about the origin

5 c)

$$iz = 1 + 3i$$

Rotation by 90° anticlockwise about the origin (or 270° clockwise)

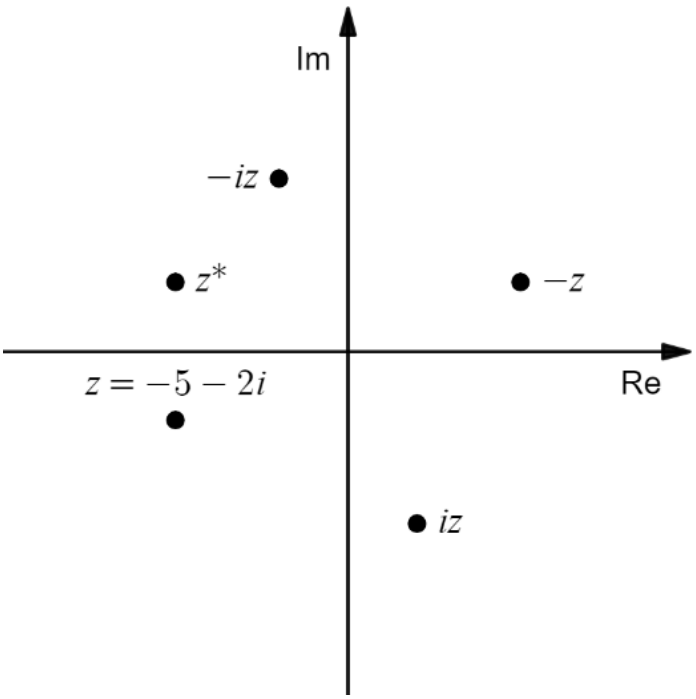
5 d)

$$-iz = -1 - 3i$$

Rotation by 270° anticlockwise about the origin (or 90° clockwise)

6)

$z = -5 - 2i$, so that $z^* = -5 + 2i$, $-z = 5 + 2i$, $iz = 2 - 5i$, $-iz = -2 + 5i$



6 a)

$$z^* = -5 + 2i$$

Reflection across the real axis

6 b)

$$-z = 5 + 2i$$

Rotation by 180° about the origin

6 c)

$$iz = 2 - 5i$$

Rotation by 90° anticlockwise about the origin (or 270° clockwise)

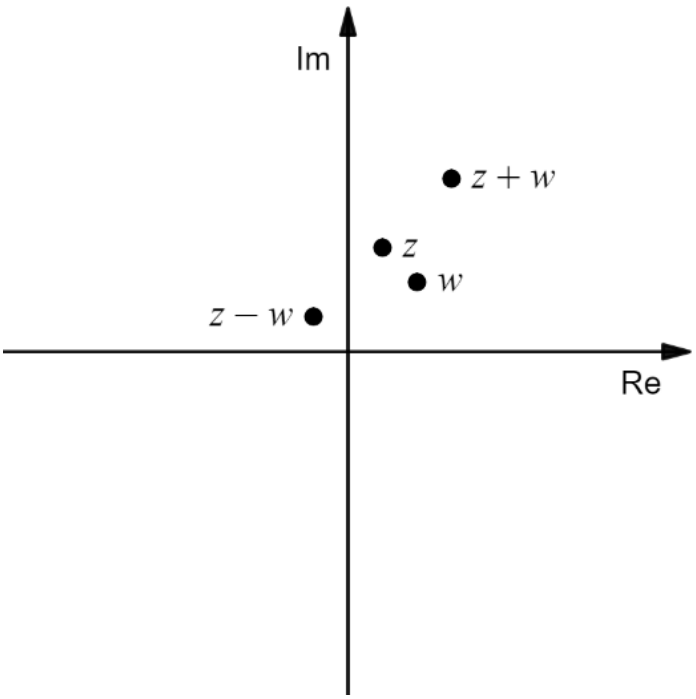
6 d)

$$-iz = -2 + 5i$$

Rotation by 270° anticlockwise about the origin (or 90° clockwise)

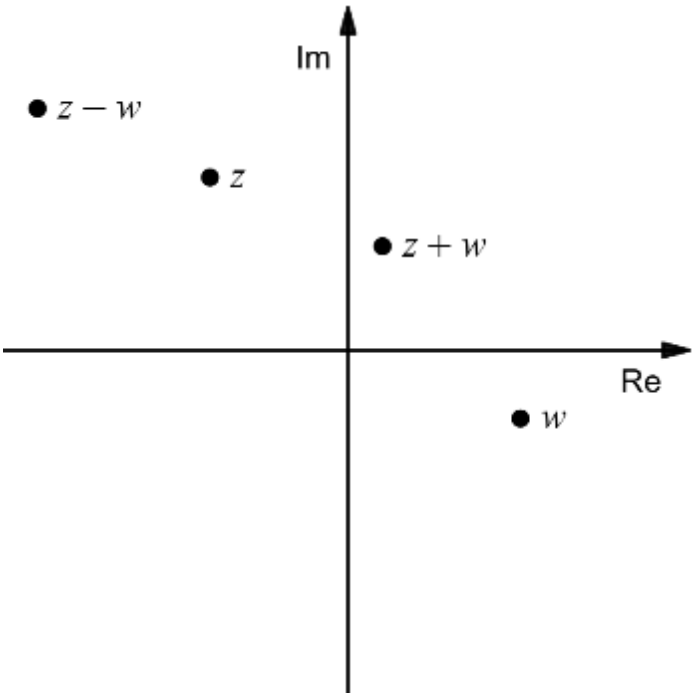
7)

$z + w = 3 + 5i$ and $z - w = -1 + i$



8)

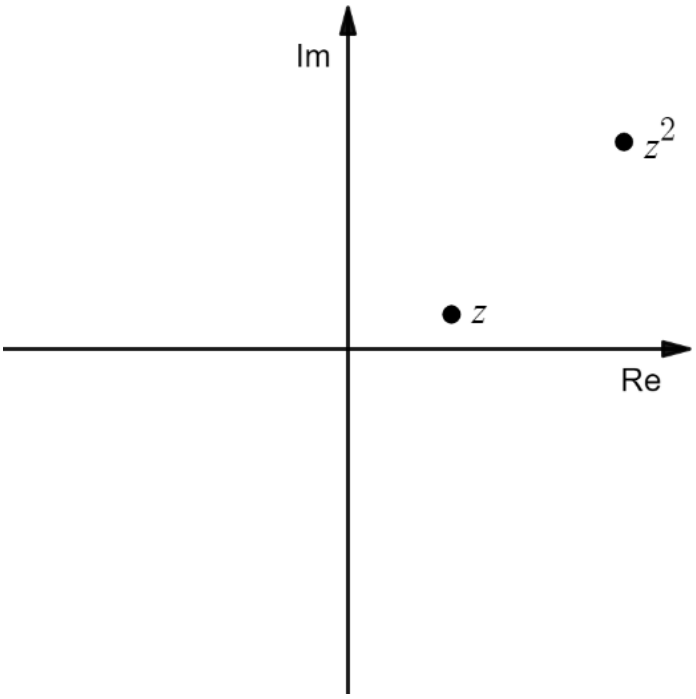
$z+w=1+3i$ and $z-w=-9+i$



9 a)

$$z^2=(3+i)(3+i)=9+6i-1=8+6i \text{ as req.}$$

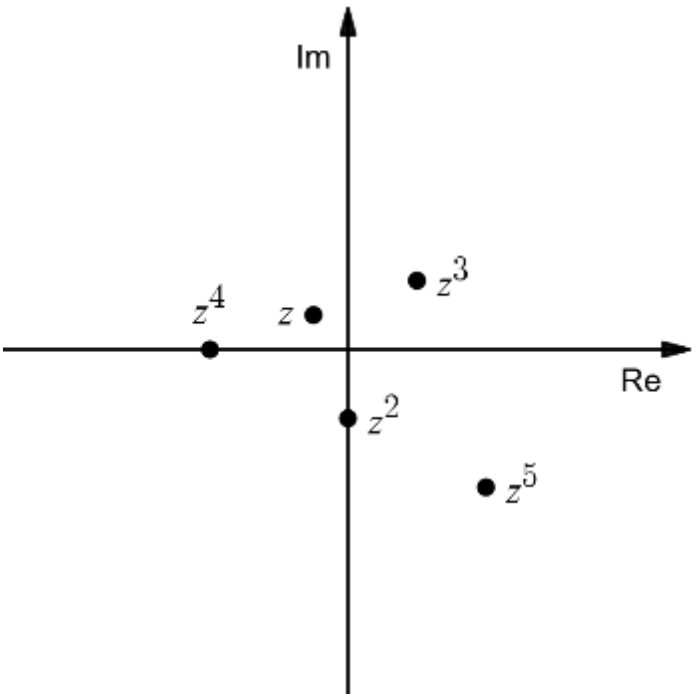
9 b)



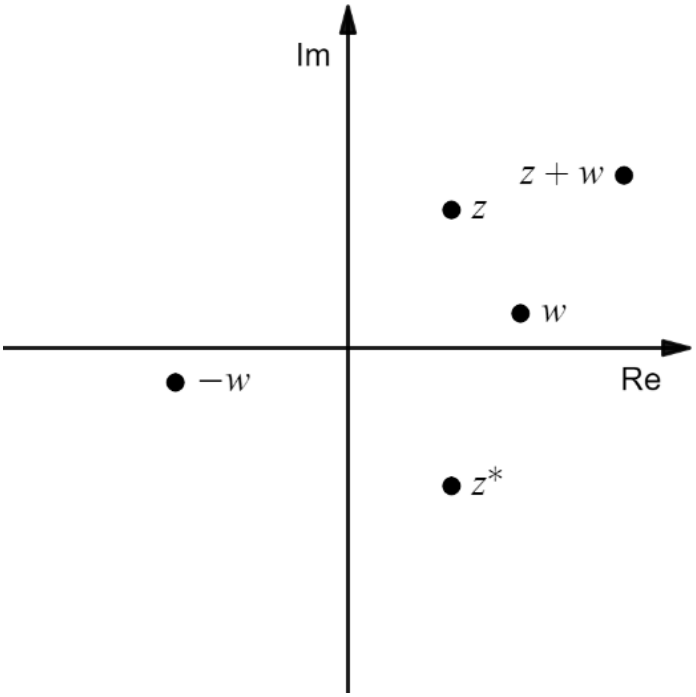
10 a)

$$z^2 = (-1+i)(-1+i) = 1 - 2i - 1 = -2i$$
$$z^3 = (-1+i)(-2i) = 2 + 2i$$
$$z^4 = (-1+i)(2+2i) = -2 - 2i + 2i - 2 = -4$$
$$z^5 = -4(-1+i) = 4 - 4i$$

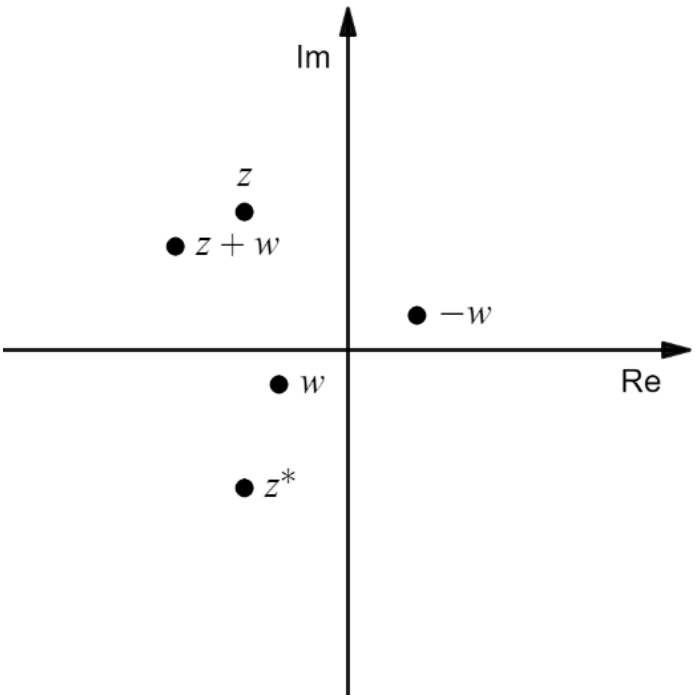
10 b)



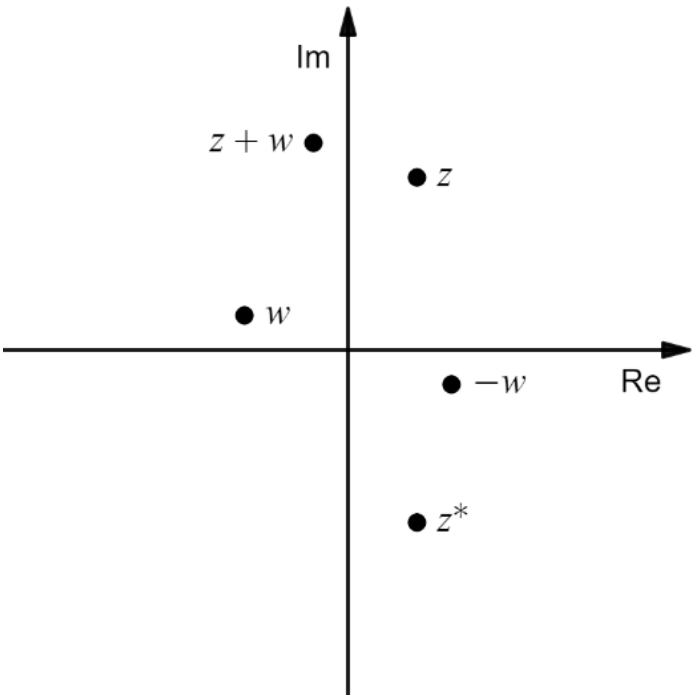
11 a)



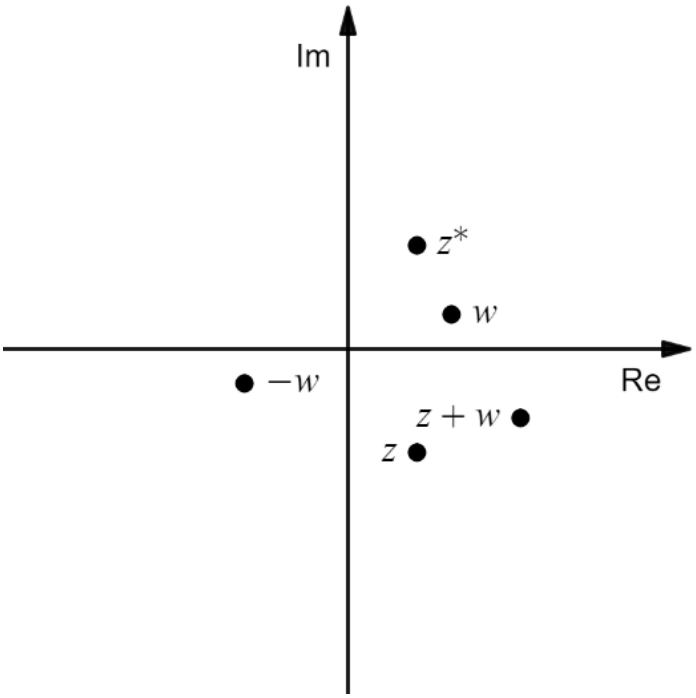
11 b)



11 c)



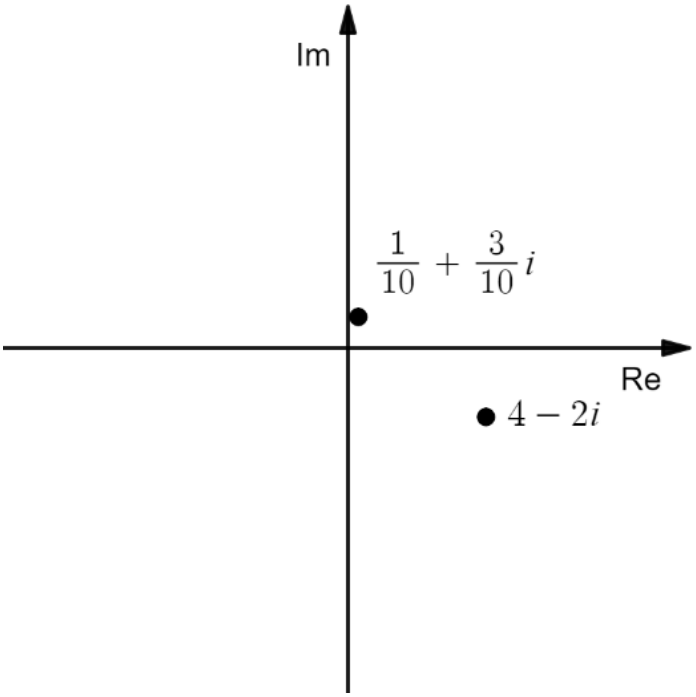
11 d)



12 a)

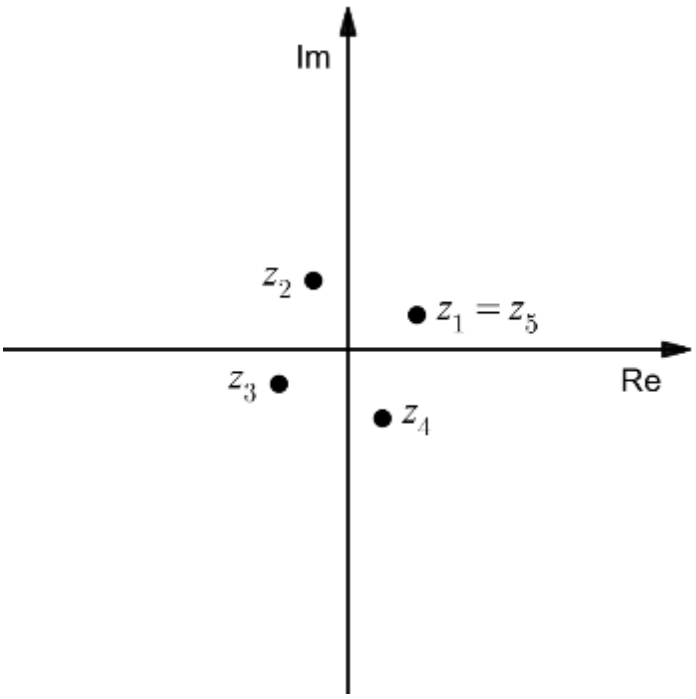
$$z_2 = \frac{1+i}{4-2i} \times \frac{4+2i}{4+2i} = \frac{4+6i-2}{16+4} = \frac{2+6i}{20} = \frac{1}{10} + \frac{3}{10}i$$

12 b)



13 a)

$$z_2 = i(2+i) = -1+2i$$
$$z_3 = -(2+i) = -2-i$$
$$z_4 = -i(2+i) = 1-2i$$
$$z_5 = 1(2+i) = 2+i$$

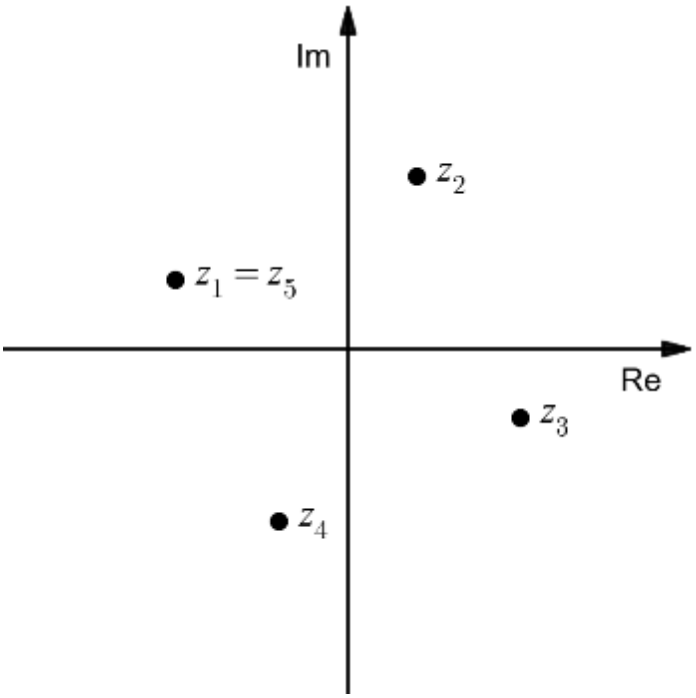


13 b)

Rotation of 90° anti-clockwise (or 270° clockwise) about the origin.

14 a)

$$z_2 = -i(-5+2i) = 2+5i$$
$$z_3 = -i(2+5i) = 5-2i$$
$$z_4 = -i(5-2i) = -2-5i$$
$$z_5 = -i(-2-5i) = -5+2i$$

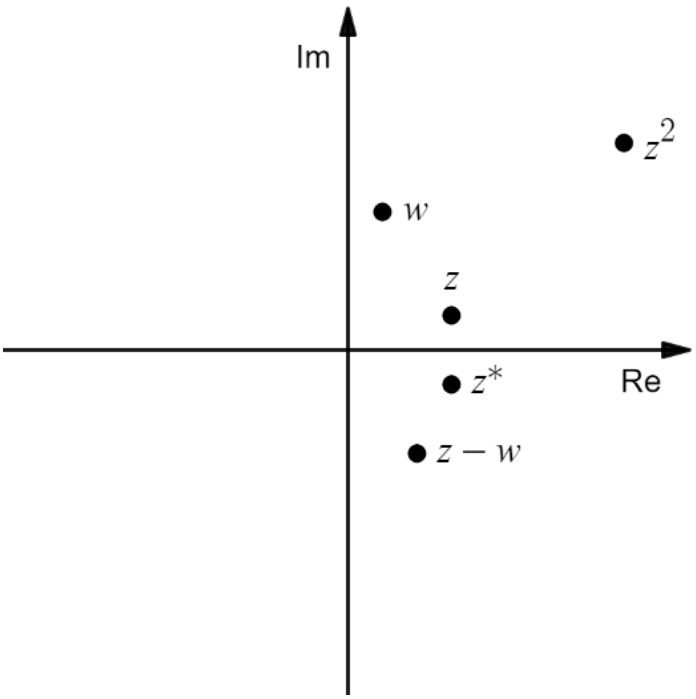


14 b)

Rotation by 90° clockwise (or 270° anti-clockwise) about the origin

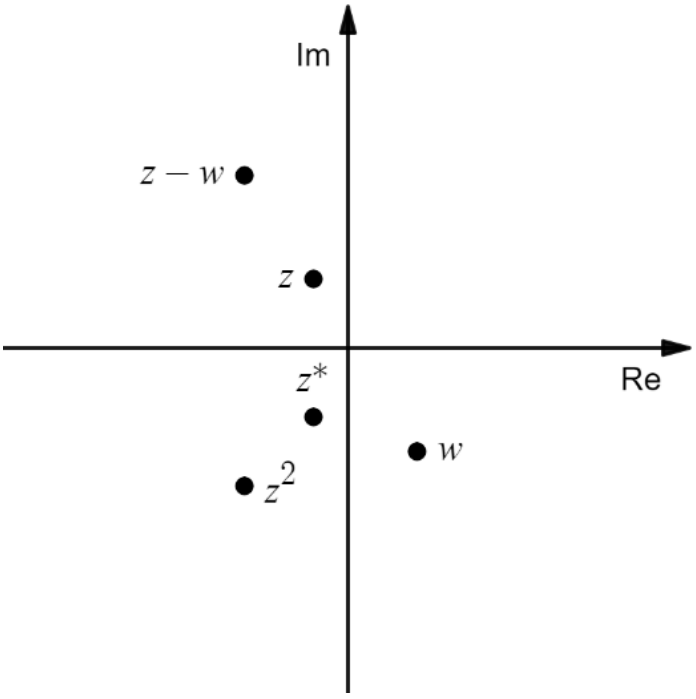
15 a)

$(a+bi)^2 = a^2 - b^2 + 2abi$ since $a > b$, z^2 is in top right quadrant.

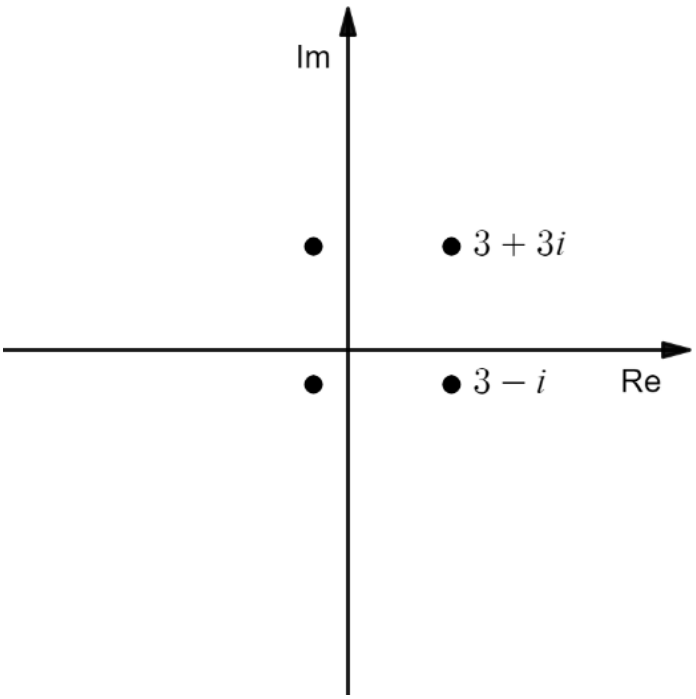


15 b)

$(-a + bi)^2 = a^2 - b^2 - 2abi$ since $b > a$, z^2 is in bottom left quadrant.



16 a)



The area is $4 \times 4 = 16$ units^2

16 b)

$-1 + 3i$ and $-1 - i$ or $7 - i$ and $7 + 3i$

17 a)

$$r = \sqrt{a^2 + b^2}$$

17 b)

$$zz^* = (a + bi)(a - bi) = a^2 + b^2 = r^2 \text{ as req.}$$

18 a)

$$(a + bi)(c + di) = ac - bd + (bc + ad)i$$

18 b)

$p = \sqrt{a^2 + b^2}$ and $q = \sqrt{c^2 + d^2}$

$$\begin{aligned} \text{so } pq &= \sqrt{(a^2 + b^2)(c^2 + d^2)} \\ &= \sqrt{a^2c^2 + b^2d^2 + b^2c^2 + a^2d^2} \\ |zw| &= \sqrt{(ac - bd)^2 + (bc + ad)^2} \\ &= \sqrt{a^2c^2 - 2abcd + b^2d^2 + b^2c^2 + 2abcd + a^2d^2} \\ &= \sqrt{a^2c^2 + b^2d^2 + b^2c^2 + a^2d^2} \\ \therefore |zw| &= pq \text{ as req.} \end{aligned}$$

19 a)

$$iz = i(2 - 3i) = 3 + 2i$$

19 b)

$$(1 + i)z = (1 + i)(2 - 3i) = 2 - 3i + 2i + 3 = 5 - i$$

19 c)

Distance between each of the points:

$$\begin{aligned}
|z - 0| &= |2 - 3i| = \sqrt{2^2 + (-3)^2} = \sqrt{13} \\
|0 - iz| &= |0 - 3 - 2i| = |-3 - 2i| = \sqrt{(-3)^2 + (-2)^2} = \sqrt{13} \\
|iz - (1 + i)z| &= |3 + 2i - 5 + i| = |-2 + 3i| = \sqrt{(-2)^2 + 3^2} = \sqrt{13} \\
|(1 + i)z - z| &= |5 - i - 2 + 3i| = |3 + 2i| = \sqrt{3^2 + 2^2} = \sqrt{13}
\end{aligned}$$

All the lengths are the same.

Since the transformation from z to iz is a rotation of 90° anti-clockwise about the origin, one of the angles must be 90° so it must be a square.

19 d)

$\sqrt{13} \times \sqrt{13} = 13$ units^2

20 a)

$$iz = i(a + bi) = -b + ai$$

20 b)

$$(1 + i)z = (1 + i)(a + bi) = a + bi + ai - b = (a - b) + (a + b)i$$

20 c)

Distance between each of the points:

$$\begin{aligned}
|z - 0| &= |a + bi| = \sqrt{a^2 + b^2} \\
|0 - iz| &= |0 + b - ai| = |b - ai| = \sqrt{b^2 + (-a)^2} = \sqrt{a^2 + b^2} \\
|-b + ai - (a - b) - (a + b)i| &= |-b + ai - a + b - ai - bi| \\
&= |-a - bi| = \sqrt{(-a)^2 + (-b)^2} = \sqrt{a^2 + b^2} \\
|(a - b) + (a + b)i - a - bi| &= |a - b + ai + bi - a - bi| = |-b + ai| \\
&= \sqrt{(-b)^2 + a^2} = \sqrt{a^2 + b^2}
\end{aligned}$$

All the lengths are the same.

Since the transformation from z to iz is a rotation of 90° anti-clockwise about the origin, one of the angles must be 90° so it must be a square.

20 d)

$\sqrt{a^2+b^2} \times \sqrt{a^2+b^2} = (a^2+b^2)$ units^2

21 a)

$$iz = i(a+bi) = -b+ai$$

So $|OP| = \sqrt{a^2+b^2}$ and $|OQ| = \sqrt{(-b)^2+a^2} = \sqrt{a^2+b^2}$

21 b)

Geometrically, multiplying by i rotates a complex number 90° anti-clockwise about the origin on the Argand diagram so OP is perpendicular to OQ

Alternative Method:

gradient of line from origin to z is $\frac{b}{a}$

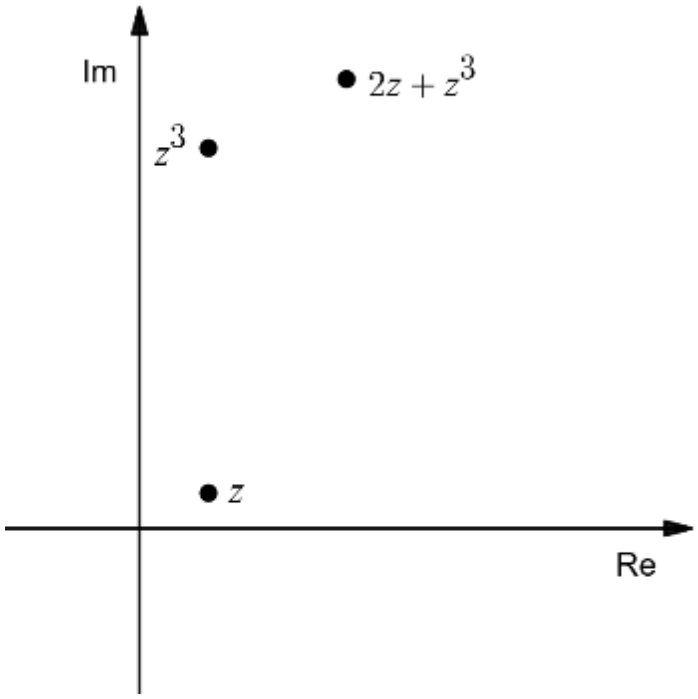
gradient of line from origin to $iz = -b+ai$ is $\frac{a}{-b}$

Since $\frac{b}{a} \times \frac{a}{-b} = -1$ they are perpendicular, so OP is perpendicular to OQ

22 a)

$z = 2+i$ gives $z^3 = (2+i)^3 = 8+12i-6-i = 2+11i$

$$2z+z^3 = 2(2+i)+2+11i = 6+13i$$



22 b)

$$|OA| = \sqrt{2^2+1^2} = \sqrt{5}$$

22 c)

$$|BC| = |6 + 13i - 2 - 11i| = |4 + 2i| = \sqrt{4^2 + 2^2} = \sqrt{20} = 2\sqrt{5}$$

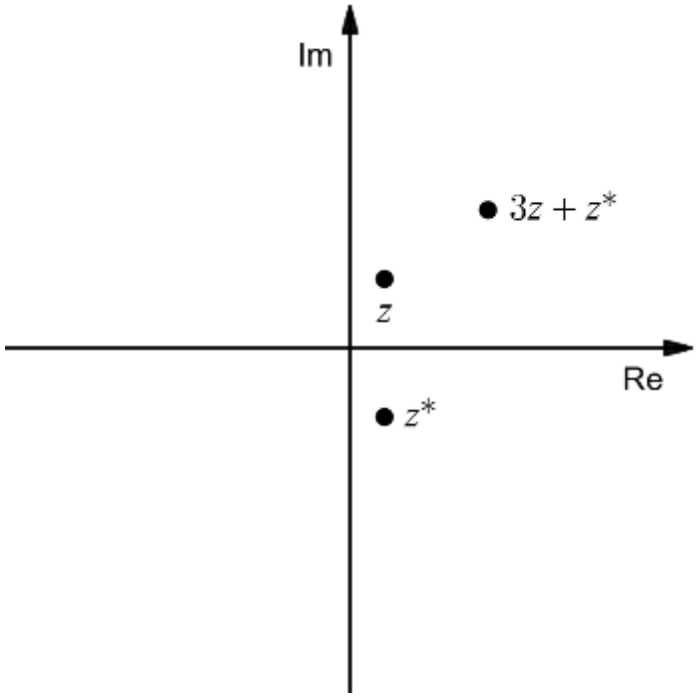
22 d)

Line segment $|BC|$ is twice the length of $|OA|$

23 a)

$z = 1 + 2i$ gives $z^* = 1 - 2i$

$$3z + z^* = 3(1 + 2i) + 1 - 2i = 4 + 4i$$



23 b)

$$|AC| = \sqrt{3^2 + 4^2} = \sqrt{13}$$

23 c)

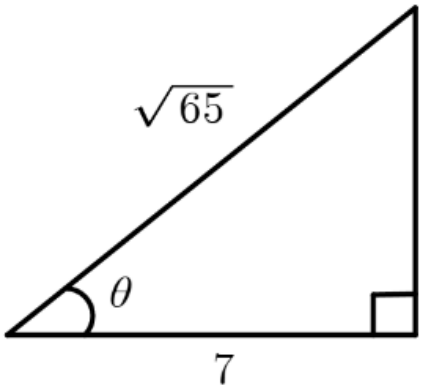
$$|BC| = |4 + 4i - 1 + 2i| = |3 + 6i| = \sqrt{3^2 + 6^2} = \sqrt{45} = 3\sqrt{5}$$

23 d)

Area of $ABC = \frac{1}{2} \times 4 \times 3 = 6$ units^2

Alternative Method:

$$\begin{aligned} |AB| &= 4 \\ 4^2 &= (3\sqrt{5})^2 + (\sqrt{13})^2 - 2 \times (3\sqrt{5})(\sqrt{13}) \cos\theta \\ 16 &= 45 + 13 - 6\sqrt{65} \cos\theta \\ \cos\theta &= \frac{7}{\sqrt{65}} \end{aligned}$$



So $\sin\theta = \frac{4}{\sqrt{65}}$

Area of $ABC = \frac{1}{2} \times 3\sqrt{5} \times \sqrt{13} \times \frac{4}{\sqrt{65}} = 6$ units^2

Alternative Method:

$$|AB| = 4$$

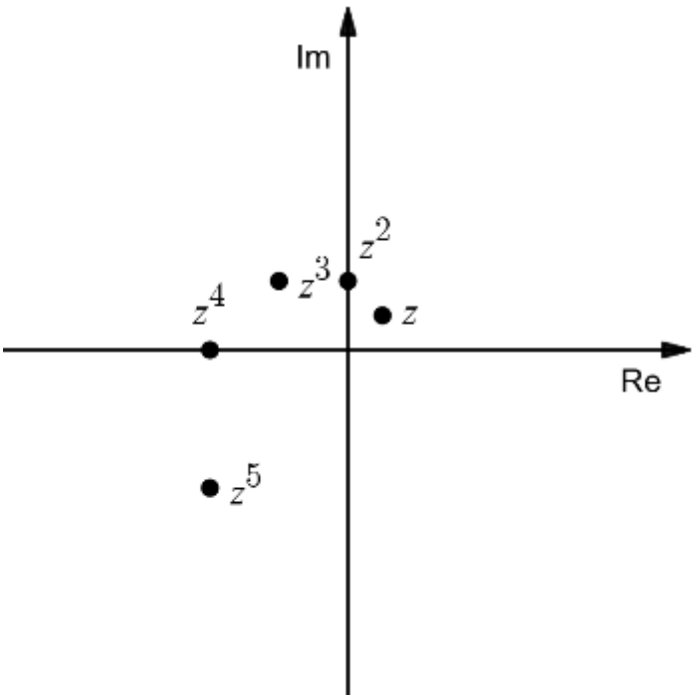
$$s = \frac{4 + 3\sqrt{5} + \sqrt{13}}{2}$$

$$\text{Area} = \sqrt{\frac{4+3\sqrt{5}+\sqrt{13}}{2}\left(\frac{4+3\sqrt{5}+\sqrt{13}}{2} - 3\sqrt{5}\right)\left(\frac{4+3\sqrt{5}+\sqrt{13}}{2} - \sqrt{13}\right)\left(\frac{4+3\sqrt{5}+\sqrt{13}}{2} - 4\right)} = 6 \text{ units}^2$$

24 a)

$$\begin{aligned} z &= 1 + i \\ z^2 &= (1+i)(1+i) = 1 + i + i - 1 = 2i \\ z^3 &= 2i(1+i) = -2 + 2i \\ z^4 &= (2i-2)(1+i) = 2i - 2 - 2 - 2i = -4 \\ z^5 &= -4(1+i) = -4 - 4i \end{aligned}$$

24 b)



24 c)

$$|z| = |1+i| = \sqrt{1^2+1^2} = \sqrt{2}$$
$$|z^2| = |2i| = 2$$
$$|z^3| = |-2+2i| = \sqrt{(-2)^2+2^2} = \sqrt{8} = 2\sqrt{2}$$
$$|z^4| = |-4| = 4$$
$$|z^5| = |-4-4i| = \sqrt{(-4)^2+(-4)^2} = \sqrt{32} = 4\sqrt{2}$$

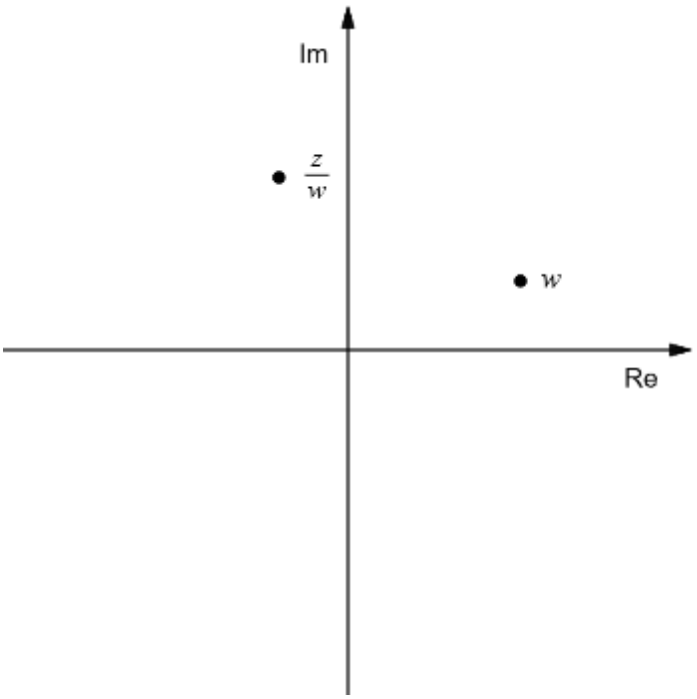
24 d)

$$|z^n| = (\sqrt{2})^n = 2^{\frac{n}{2}}$$

25 a)

$$\frac{z}{w} = \frac{-20+21i}{5+2i} \times \frac{5-2i}{5-2i} = \frac{-58+145i}{25+4} = -2+5i \text{ as req.}$$

25 b)



25 c)

$w = 5 + 2i$ and $\frac{z}{w} = -2 + 5i$

Since $wi = i(5 + 2i) = -2 + 5i = z$

Geometrically, multiplying by i rotates a complex number 90° anti-clockwise about the origin on the Argand diagram so they are perpendicular, the points w and $\frac{z}{w}$ form a right-angled triangle.

Alternative Method:

gradient of line from origin to w is $\frac{5}{2}$

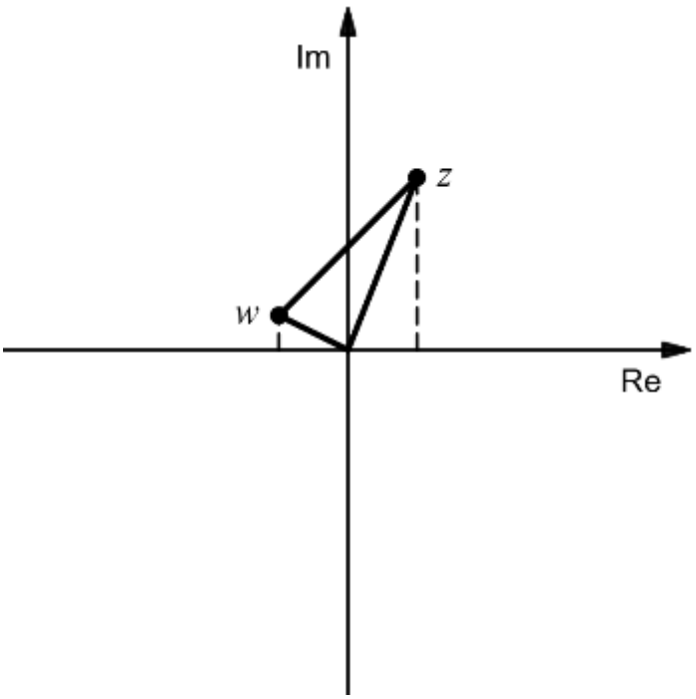
gradient of line from origin to $\frac{z}{w}$ is $\frac{-2}{5}$

Since $\frac{5}{2} \times \frac{-2}{5} = -1$ they are perpendicular, the points w and $\frac{z}{w}$ form a right-angled triangle.

25 d)

Area $= \frac{1}{2}\sqrt{(-2)^2 + 5^2}\sqrt{5^2 + 2^2} = \frac{29}{2}$ units^2

26)



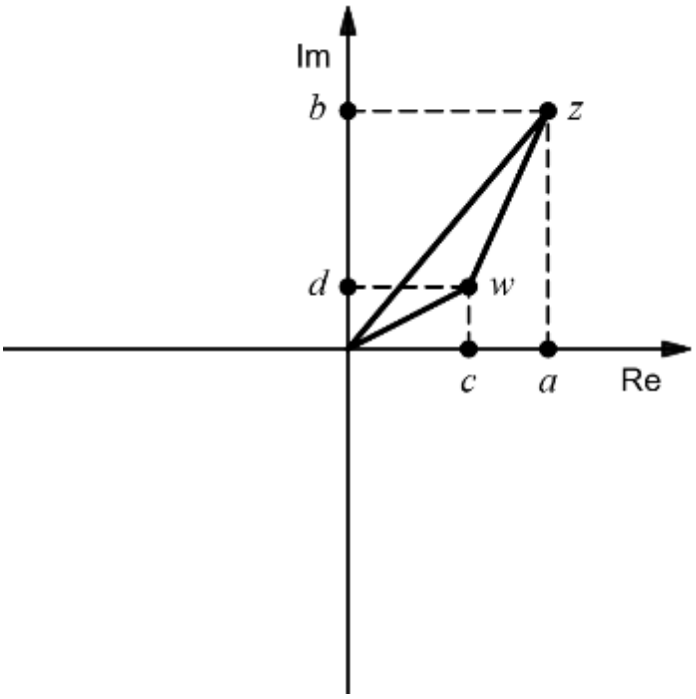
Area of trapezium: $\frac{1}{2}(1+5)\times 4 = 12$

Area of triangle 1: $\frac{1}{2}\times 2\times 1 = 1$

Area of triangle 2: $\frac{1}{2}\times 2\times 5 = 5$

$\therefore$ Area is $12-1-5 = 6$ units^2 as req.

27)



Large triangle: $\frac{1}{2}ab$ and small triangle: $\frac{1}{2}cd$

Trapezium: $\frac{1}{2}(b+d)(a-c)$

So area wanted:

$$\begin{aligned}
&\frac{1}{2}ab - \frac{1}{2}cd - \frac{1}{2}(b+d)(a-c) \\
&= \frac{1}{2}ab - \frac{1}{2}cd - \frac{1}{2}(ba - bc + da - dc) \\
&= \frac{1}{2}ab - \frac{1}{2}cd - \frac{1}{2}ab + \frac{1}{2}cb - \frac{1}{2}ad + \frac{1}{2}cd \\
&= \frac{1}{2}bc - \frac{1}{2}ad \\
&= \frac{1}{2}(bc - ad)
\end{aligned}$$

To keep the area positive: $\frac{1}{2}|bc - ad|$

But since $zw^* = (a+bi)(c-di) = ac + bd + (bc - ad)i$

$$\mathrm{Im}(zw^*) = bc - ad$$

So that area $= \frac{1}{2}|\,\mathrm{Im}(zw^*)|$ as req.

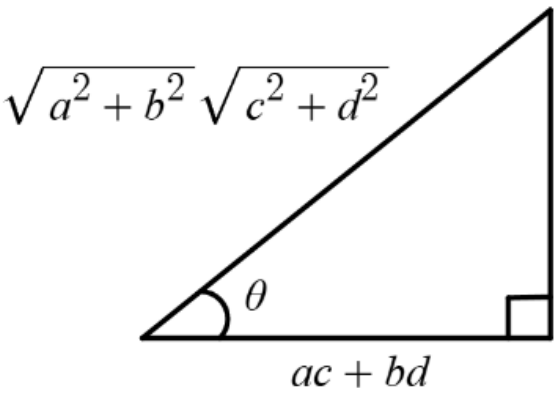
Alternative Method:

Let $z = a + bi$ and $w = c + di$

$zw^* = (a+bi)(c-di) = ac + bd + (bc - ad)i$ so $\mathrm{Im}(zw^*) = bc - ad$

$|z^*| = \sqrt{a^2 + b^2}$ and $|w| = \sqrt{c^2 + d^2}$

$$\cos\theta = \frac{ac + bd}{\sqrt{a^2 + b^2}\sqrt{c^2 + d^2}}$$



So that the opposite side is given by:

$$\begin{aligned}
&\sqrt{(a^2+b^2)(c^2+d^2) - (ac+bd)^2} \\
&= \sqrt{a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2 - a^2c^2 - 2abcd - b^2d^2} \\
&= \sqrt{a^2d^2 + b^2c^2 - 2abcd} \\
&= \sqrt{(ad - bc)^2} \\
&= ad - bc
\end{aligned}$$

$$\sin\theta = \frac{ad - bc}{\sqrt{a^2 + b^2}\sqrt{c^2 + d^2}}$$

$$\begin{aligned}
\text{Area: } & \frac{1}{2}\sqrt{a^2+b^2}\sqrt{c^2+d^2} \times \frac{ad-bc}{\sqrt{a^2+b^2}\sqrt{c^2+d^2}} \\
&= \frac{1}{2}(ad-bc) \\
&= \frac{1}{2}|ad-bc| \\
&= \frac{1}{2}|bc-ad| \\
&= \frac{1}{2}|\operatorname{Im}(zw^*)| \text{ as req.}
\end{aligned}$$